

THONG PHAPHUM, Thailand

WMO: 484210

Lat: 14.742N

Lon: 98.636E

Elev: 99

StdP: 100.14

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.1	16.6	11.7	8.7	22.4	13.3	9.6	23.5	4.5	29.6	3.6	29.9	0.0	N/A	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.4	38.6	25.7	37.5	25.2	36.5	24.9	27.6	34.0	27.2	33.0	27.0	32.5	1.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.9	30.0	25.9	21.5	29.5	25.7	21.2	29.2	88.7	34.1	86.9	33.4	85.5	32.5	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.5	3.1	2.6	DB	12.5	40.0	2.2	1.2	10.9	40.9	9.7	41.5	8.4	42.2	6.9	43.0	(4)
(5)			WB	11.8	29.2	2.1	1.3	10.3	30.1	9.1	30.8	7.9	31.5	6.3	32.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	25.8	27.7	29.5	30.6	29.3	28.1	27.3	27.3	27.8	27.9	26.8	25.3	(6)
(7)		DBStd	2.15	2.05	1.81	1.51	1.56	1.94	1.33	1.16	1.17	1.11	1.13	1.62	2.24	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6490	489	496	604	618	600	542	538	536	534	554	504	475	(10)
(11)		CDD18.3	3448	231	262	346	368	342	292	279	277	284	296	254	217	(11)
(12)		CDH23.3	35284	2217	2969	4299	4684	3809	2894	2467	2386	2544	2753	2312	1949	(12)
(13)		CDH26.7	15750	1069	1626	2487	2664	1735	991	708	661	846	1078	1016	869	(13)
(14)	Wind	WSAvg	0.2	0.2	0.4	0.5	0.4	0.2	0.1	0.1	0.2	0.2	0.1	0.2	0.3	(14)
(15)	Precipitation	PrecAvg	2295	7	12	57	98	264	367	480	476	329	177	24	5	(15)
(16)		PrecMax	2802	34	75	194	360	456	683	895	792	564	331	92	42	(16)
(17)		PrecMin	1543	0	0	0	10	91	241	177	170	175	25	0	0	(17)
(18)		PrecStd	323	10	19	45	69	92	94	157	121	96	70	24	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	37.4	39.1	40.2	39.5	35.7	34.0	33.8	34.2	34.5	35.0	34.7	(19)
(20)			MCWB	23.9	23.8	25.0	26.2	26.7	26.8	26.6	26.5	26.8	26.5	26.0	25.0	(20)
(21)		2%	DB	34.4	36.4	38.0	38.9	37.5	33.9	32.6	32.5	33.2	33.6	34.0	33.7	(21)
(22)			MCWB	23.7	23.6	24.5	26.1	26.5	26.6	26.4	26.3	26.5	26.4	25.6	24.4	(22)
(23)		5%	DB	33.4	35.6	37.2	37.8	35.6	32.5	31.3	31.3	32.2	32.9	32.9	32.6	(23)
(24)			MCWB	23.4	23.5	24.2	25.9	26.6	26.5	26.2	26.1	26.4	26.2	25.1	23.8	(24)
(25)		10%	DB	32.3	34.5	36.2	36.6	33.8	31.2	30.2	30.1	31.0	31.9	31.9	31.5	(25)
(26)			MCWB	23.0	23.2	24.1	25.8	26.5	26.3	26.0	25.9	26.2	26.0	24.7	23.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	26.2	27.2	28.5	28.3	27.7	27.2	27.2	27.5	27.6	27.2	26.5	(27)
(28)			MADB	32.3	34.0	36.2	37.4	35.3	32.5	31.6	31.6	32.3	32.7	32.9	32.3	(28)
(29)		2%	WB	25.1	25.5	26.3	27.5	27.5	27.2	26.8	26.7	27.0	27.1	26.6	25.6	(29)
(30)			MADB	31.4	33.0	34.8	35.9	34.0	31.7	30.8	30.8	31.7	32.2	32.0	31.5	(30)
(31)		5%	WB	24.5	24.9	25.8	27.0	27.1	26.8	26.5	26.4	26.6	26.7	26.0	24.9	(31)
(32)			MADB	30.7	32.4	33.9	34.9	33.1	31.1	30.2	30.1	31.0	31.5	31.3	30.7	(32)
(33)		10%	WB	23.9	24.4	25.3	26.5	26.7	26.5	26.1	26.1	26.3	26.3	25.5	24.3	(33)
(34)			MADB	30.0	31.6	32.9	34.0	32.1	30.4	29.6	29.4	30.2	30.7	30.1	29.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.2	14.8	14.1	12.4	8.7	6.7	6.1	7.1	8.6	10.9	12.9	(35)	
(36)			MCDBR	14.7	15.5	15.5	14.2	11.9	8.8	8.0	8.1	8.8	9.9	11.8	13.3	(36)
(37)			MCWBR	5.3	4.7	4.5	3.9	3.0	2.8	2.9	2.8	3.1	3.6	4.3	5.0	(37)
(38)		5% WB	MCDBR	12.7	13.3	13.4	12.5	9.8	7.5	7.0	7.0	8.0	8.9	10.5	11.9	(38)
(39)			MCWBR	4.8	4.4	4.2	4.0	3.1	2.8	2.9	2.8	3.1	3.5	4.1	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.499	0.587	0.672	0.577	0.484	0.448	0.443	0.443	0.448	0.537	0.482	0.477	(40)	
(41)		taud	2.060	1.858	1.687	1.962	2.269	2.386	2.398	2.386	2.361	2.058	2.190	2.147	(41)	
(42)		Ebn at Noon	793	741	691	760	822	843	849	858	858	772	805	802	(42)	
(43)		Edh at Noon	162	206	251	190	138	121	120	123	126	167	141	145	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.84	5.34	5.63	5.94	4.78	4.08	3.71	3.72	4.16	4.60	4.83	4.74	(44)	
(45)		RadStd	0.23	0.23	0.39	0.36	0.53	0.49	0.47	0.40	0.39	0.29	0.28	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.21	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	+0.33	+0.35	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page